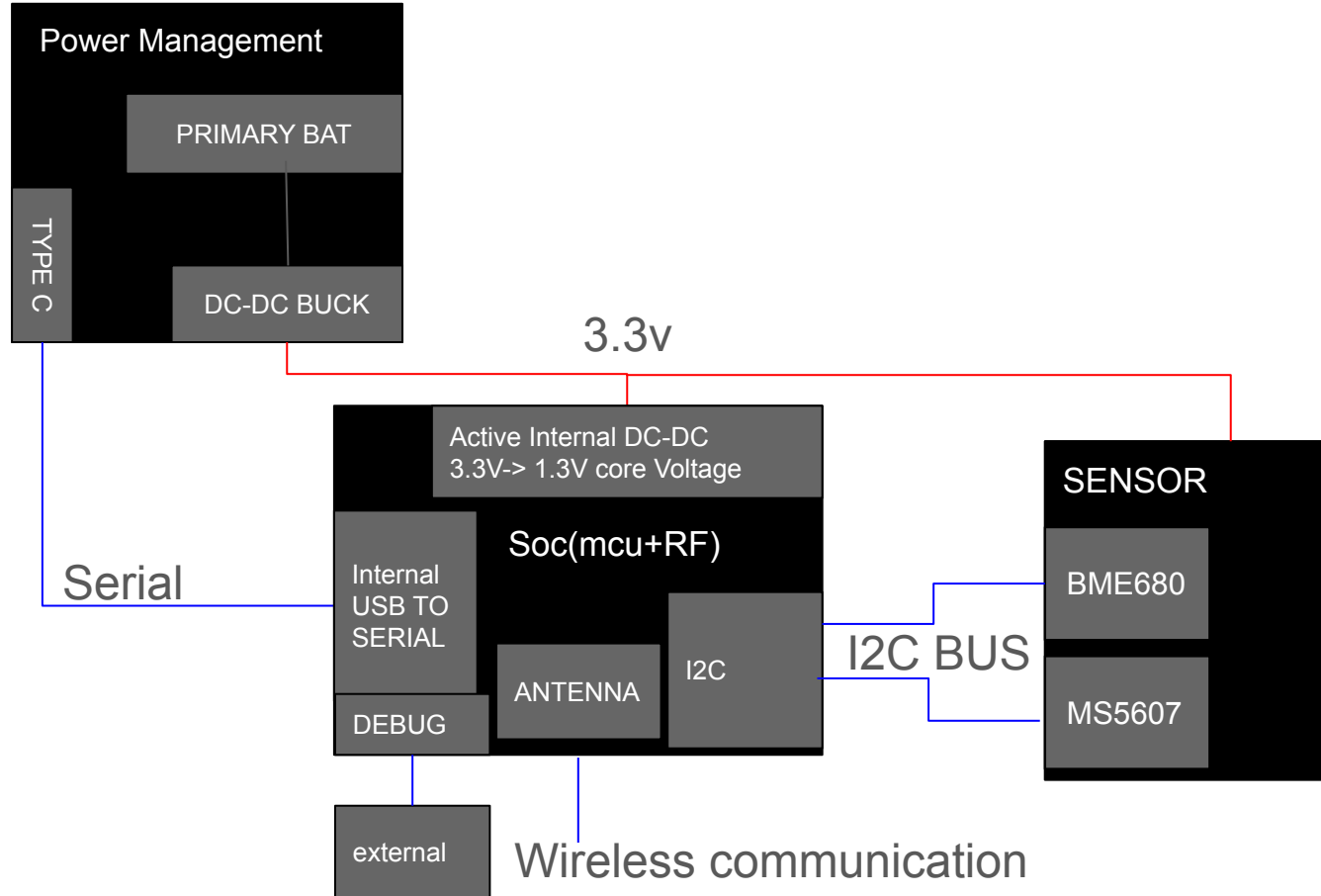


Block Diagram



Energy logic & Analysis

- battery capacity : 1200 mAh
- sleep current: 36uA (fixed cost)
- Annuary sleep energy consumption : $0.0036 \text{ mA} \times 8760 \text{ hours} = 31.536 \text{ mAh}$
- Remaining capacity for measurement :

$$1,020 \text{ mAh} - 31.536 = 988.464 \text{ mAh}$$

The Active measurement (1 cycle)

	MCU	ms5607	BMR680	BLE			
	Wake-Up	T _{sensor1}	T _{sensor2}	T _{RF}	Deep sleep (ms)	T _{period}	1 Cycle
Duration (ms)	2.5	18.08	245.00	4	299,730.42	269.58	300,000
Current (mA)	4.8	6.2	13.80	8			
Battery Reffered Current(mA)	4.368	5.642	12.56	7.28			
Energy Consumption(mAs)	10.92	112.096	3,381.00	32		3,536.02	

heating time included

0.26 seconds

- current consumption per measurement : 0.002456 mAh
 active current : 32.8 mA
 duration : 269.58 ms